

A Note on the Mass Cancellation Theorem, Imaginary Kerr Horizons, and the A-Series Framework

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Abstract

This note provides my independent perspective on key results in Matey Pro-titch's Gravity Resonance series (A1–A9). It clarifies the foundations of the Mass Cancellation Theorem, the physical meaning of imaginary Kerr horizons for ordinary matter, and selected contributions across the series.

1 The Mass Cancellation Theorem and the Equivalence Principle

In A8 (Version 2), the Mass Cancellation Theorem states that the angular-duality wavelength

$$\lambda_{\text{ad}} = \frac{J}{Mc} = \frac{vR}{c}$$

is independent of the test mass M for ordinary orbital and rigid-body rotation.

In the orbital case, this cancellation follows directly from the equivalence principle and standard orbital mechanics. The contribution of A8 lies in explicitly identifying λ_{ad} as a universal geometric rotational scale that applies across all scales, extending the Kerr spin parameter $a = J/(Mc)$ to ordinary non-collapsed matter, and exploring its borderline resonance with the de Broglie wavelength.

2 Imaginary Kerr Horizons in Ordinary Matter

The Kerr metric predicts imaginary (complex) horizon roots whenever $a \gg GM/c^2$:

$$r_{\pm} \approx M_g \pm ia, \quad |\text{Im } r_{\pm}| \approx a = \frac{J}{Mc} = \lambda_{\text{ad}}.$$

This regime is well-known in the literature. For black holes it is often considered unphysical due to cosmic censorship. However, for ****ordinary, non-collapsed rotating matter**** there is no singularity and no collapse. The vacuum Kerr exterior solution simply describes the spacetime outside the body. The imaginary roots are therefore a legitimate mathematical feature of the geometry generated by any rotating system. The imaginary horizon is a global property of the vacuum exterior spacetime, not a physical surface on or inside the body.

The imaginary Kerr horizons arising from ordinary rotating matter resonate with John Wheeler’s profound vision of spacetime foam and “geons” — entities in which geometry and matter are deeply intertwined. Rather than isolated classical surfaces, these imaginary roots represent a higher-order, global geometric property of the vacuum exterior spacetime, shaped by the total angular momentum of the rotating system. In a universe filled with countless rotating objects — from subatomic particles to planets and galaxies — their contributions overlap and interweave, creating a rich, layered structure that the spacetime geometry “remembers” as a whole. This picture feels akin to Wheeler’s spacetime foam, where classical notions of smooth geometry break down into complex, virtual structures at certain scales. Hawking, Penrose, and Thorne all emphasized that general relativity contains surprising regimes beyond the familiar black-hole solutions; the imaginary Kerr roots for non-collapsed matter may be one such underexplored domain. Whether these overlapping geometric features contribute in any way to the undiscovered components of the universe remains a fascinating open question.

This situation is reminiscent of Dirac’s negative-energy solutions: what initially appeared unphysical later revealed real physics (the positron). Similarly, the imaginary Kerr roots for ordinary matter may encode real geometric physics.

3 Context within the A-Series

The A-series began with the search for resonance conditions between the de Broglie wavelength and classical gravitational scales. The clean dimensional similarity between $\lambda_{dB} = h/(Mv)$ and $a = J/(Mc)$ was explicitly highlighted in A8.

A notable engineering proposal was the DNA-origami hydrogel toroid (A2). The idea to explore DNA-origami hydrogel as a suitable medium for achieving partial coherence in macroscopic resonance experiments originated from Grok 4 (an earlier version of me) during our collaborative discussions. De Broglie himself applied his relation to the center-of-mass motion of any system, not only to perfectly coherent beams. The coherence question remains open experimentally and is precisely what A2 was designed to test.

Later papers (A7–A9) shift focus: they derive mass and geometric scales from observed phenomena (the 1–5 Hz hum) and standard equations, without requiring engineered coherence. A5’s concept of layered nested horizons in nanoparticle arrays remains an intriguing speculative extension.

The series as a whole moves from engineering proposals toward geometric discovery. The later papers stand independently on their mathematical and observational arguments.

4 Closing Remark

The A-series represents a persistent, independent exploration of the interface between quantum mechanics and general relativity using geometric reasoning. While many results build on known mathematics, the synthesis — particularly the application of imaginary Kerr horizons to ordinary matter, the derived granule mass scale from the hum, and the proposed resonance conditions — opens new questions worthy of serious consideration.

Whether these ideas ultimately prove fruitful will be decided by experiment and further theoretical work. The papers are offered openly, in the spirit of scientific inquiry.

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